

Campaign Planning Agent.

An AI agent that automates campaign brief creation for marketing teams.

Big Data & AI in Marketing

Group Assignment

Group 1

*Option 2: Spec-Driven
Development*

The Problem

Campaign planning is often a slow, inconsistent and resource-intensive process

~4 hours

Per day spent on manual marketing tasks.

For every campaign planning cycle, marketers need to determine the channels, allocate budgets, set KPI targets and write a campaign brief.

Lack of consistency

Without dedicated media planners, two marketers working from the same data can recommend different channels, different budgets and different KPI targets → Quality varies widely

Capacity constraints

Modern campaigns run across Google Ads, Meta Ads and TikTok Ads simultaneously – each with their own metrics and pricing models.

Manual coordination raises the risk of errors and inefficiencies.

Our answer: an AI agent that does all of this in under 2 minutes.

How the Agent Works

A 6-step pipeline.

Each step makes a decision and explains why.

01 `analyze_goal()`

Funnel stage: Awareness, Consideration, or Conversion with reasoning.

02 `select_channels()`

Up to 3 channels matched to audience and funnel stage, with pricing model.

03 `distribute_budget()`

Budget split in % and amount per channel. Guaranteed to sum to 100%.

04 `calculate_kpis()`

CTR, CPA, ROAS: derived from the actual product price, not benchmarks.

05 `generate_keywords()`

Exactly 5 keywords matched to funnel stage and market language.

06 `write_brief()`

3 to 5 sentences. Ready to hand to a media buyer.

How We Built It

Spec-Driven Development: spec first, code second.

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|---|-----------------------|---|
| 1 | Write the spec | CLAUDE.md defines the agent's purpose, 6 steps, constraints, and edge cases. Written before any code. |
| 2 | Generate requirements | REQUIREMENTS.md: typed inputs, outputs, 9 constraints, 5 edge cases. |
| 3 | Generate design | DESIGN.md: file structure, function descriptions, data flow, 15 tasks in order. |
| 4 | Build step by step | Each of the 6 functions built and tested individually. |
| 5 | Test end to end | Full pipeline verified live with a real brand example and API key. |
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GlowDrop

A B2C skincare brand with 42k TikTok followers, no paid campaigns, and organic reach that is declining.

PRODUCT

Morning Boost Serum: Vitamin C face serum, 45 GBP

USP

Clean ingredients, dermatologist-tested, affordable luxury

AUDIENCE

Women 22–38, skincare and wellness, UK and US

GOAL

Increase first-time purchases

BUDGET

5,000 GBP per month

CHANNELS

Organic TikTok only (42k followers, reach declining)

The Agent Gets to Work

Step 1: analyze_goal()

Funnel Stage: Conversion

The goal “increase first-time purchases” targets a direct, measurable transactional action (a purchase), which defines the Conversion stage.

Step 2: select_channels()

Google Search (CPC)

Retargeting TikTok Ads (CPA)

Meta Ads (CPC)

Step 3: distribute_budget()

Google Search	GBP 2,500	50.0%	Primary: highest-intent channel
Retargeting TikTok Ads	GBP 1,750	35.0%	Secondary: leverages existing 42k follower base
Meta Ads	GBP 750	15.0%	Supporting: sustaining lower-intent demand

KPIs and Keywords

Step 4: All KPI targets derived from actual product price (45 GBP) : not generic benchmarks.

Channel	CTR	CPA	ROAS
Google Search	6.0%	GBP 8.50	5.29x
Retargeting TikTok Ads	3.2%	GBP 7.50	6.00x
Meta Ads	1.5%	GBP 12.50	3.6x

Step 5: `generate_keywords()`: 5 transactional keywords for Conversion stage, written in English for UK/US market.

- 1 Buy GlowDrop serum online
- 2 GlowDrop skincare first order discount
- 3 GlowDrop serum price UK
- 4 GlowDrop free trial sign up
- 5 Order GlowDrop glow serum today

The Output.

Step 6: `write_brief()` : ready to hand to a media buyer without editing.

"This conversion-stage campaign is designed to increase first-time purchases of GlowDrop serum among women aged 22 to 38 with an active interest in skincare and wellness across the UK and US. The campaign will run across Google Search, Retargeting, and Meta Ads on a total budget of GBP 5,000.00, with Google Search receiving GBP 2,500.00 (50.0%), Retargeting GBP 1,750.00 (35.0%), and Meta Ads GBP 750.00 (15.0%). Google Search should hit a 6.0% CTR, a CPA of GBP 8.50 and a 5.29x ROAS, with bidding focused on high-intent keywords including "buy GlowDrop serum online", "GlowDrop skincare first order discount", "GlowDrop serum price UK", "GlowDrop free trial sign up", and "order GlowDrop glow serum today". Retargeting must deliver a 3.2% CTR, a CPA of GBP 7.50, and a 6.00x ROAS by recapturing site visitors who did not complete a first purchase, while Meta Ads should reach a 1.5% CTR, a CPA of GBP 12.50, and a 3.60x ROAS to drive incremental new-customer acquisition. Optimise spend toward the channels meeting or exceeding these CPA and ROAS targets throughout the flight.

Business Value

2 min

vs ~4 hours

A complete media plan instead of an afternoon of manual research.

100%

budget accuracy

Percentages mathematically forced to sum to exactly 100%.

0 edits

Ready-to-use

The brief goes straight to the media buyer without changes.

***The agent does not just generate text.
It makes decisions and justifies each one based on the actual inputs not just generic advice.***

Reflection

What worked well

1. Writing the spec before any code forced us to think through every edge case upfront. Nothing was left ambiguous.
2. The agent correctly identified GlowDrop as Conversion; not Awareness; because of the 42k follower base.
3. Constraints enforced at two levels (prompt + Python) made the output reliable regardless of model behaviour.

What we would improve

1. Connect to live Google Ads and Meta API data for real CPC and CPM rates instead of benchmark ranges.
2. Add a feedback loop so the agent learns from past campaign results and improves over time.
3. Add a confidence score so users know when to question a recommendation.